

Chapter 8

Chi-Squared Tests

In Chapters 6 and 7, most of our hypothesis tests involved either a mean or a proportion. Those tests used a test statistic that measured distance from a claim in standard-error units. A large positive or negative value meant that the sample was far from what the null hypothesis predicted.

This chapter introduces a different family of questions. Instead of asking about one mean or one proportion, we ask whether a whole collection of categorical counts fits a proposed pattern. For example, we might ask whether the colors in a bag of candy match the company's stated percentages, or whether two categorical variables, such as exercise frequency and sleep quality, seem to be related.

The key idea is still familiar: compare what we *observed* to what we would *expect* if the null hypothesis were true. The difference is that we now compare many cells at once. To do this, we use the **chi-squared** statistic and the **chi-squared distribution**.

The **chi-squared distribution** (χ^2 , pronounced “ky-squared”) is not a bell curve. It is always non-negative and usually right-skewed. This makes sense because the chi-squared statistic is built from squared discrepancies. A squared discrepancy can be large or small, but it cannot be negative.

The overall logic of hypothesis testing remains the same:

1. State H_0 and H_a .
2. Compute the expected counts under H_0 .
3. Compute the chi-squared test statistic.
4. Use the chi-squared distribution to make a decision.
5. Write the conclusion in context.

The main new skill is learning how to compute expected counts and how to read the chi-squared statistic as a measure of total disagreement between observed counts and expected counts.

8.1 The Geometry of Categorical Models

Before we compute chi-squared statistics, it is helpful to step back and look at categorical data geometrically. This section is not a new procedure. Instead, it is a way to see why the procedures in this chapter make sense.

Visual Insight. A categorical probability model is a point in a shape. For two categories, the shape is a line segment. For three categories, the shape is a triangle. For four categories, the shape is a tetrahedron. The chi-squared test measures how far an observed point is from an expected-model point.

8.1.1 Categorical data as count vectors

Categorical data occur when each observation falls into one of several groups or categories. For example, suppose we ask students to choose one favorite lunch option:

Pizza, Tacos, Salad.

If 60 students answer the question, we might observe

$$O = (30, 18, 12).$$

This means that 30 students chose Pizza, 18 chose Tacos, and 12 chose Salad. The vector

$$O = (30, 18, 12)$$

is called an **observed count vector**. Each coordinate tells us how many observations landed in one category.

The total sample size is

$$30 + 18 + 12 = 60.$$

Because the total is fixed, the three numbers are connected. If the Pizza and Tacos counts are known, then the Salad count is forced:

$$\text{Salad} = 60 - \text{Pizza} - \text{Tacos}.$$

This is our first example of a **dependent parameter**: a quantity determined by the other quantities.

8.1.2 Parameters: free and dependent

A **parameter** is a number that describes a model. In categorical probability models, the parameters are usually the category probabilities.

For example, a three-category lunch model has probabilities

$$(p_1, p_2, p_3),$$

where

$$p_1 = \text{probability of Pizza}, \quad p_2 = \text{probability of Tacos}, \quad p_3 = \text{probability of Salad}.$$

Since probabilities must add to 1, we must have

$$p_1 + p_2 + p_3 = 1.$$

This equation creates a constraint. The three parameters are not completely independent.

For example, if we know

$$p_1 = 0.50 \quad \text{and} \quad p_2 = 0.30,$$

then the third probability is forced:

$$p_3 = 1 - 0.50 - 0.30 = 0.20.$$

So in a three-category model, we can think of p_1 and p_2 as **free parameters**, while p_3 is a **dependent** or **bound parameter**. It is bound by the condition that all probabilities must add to 1.

Definition: Free and dependent parameters. A **free parameter** can be chosen independently. A **dependent parameter** is determined by the free parameters and the constraints of the model.

8.1.3 Two categories: models on a line segment

Start with the simplest possible categorical model. Suppose there are only two categories:

$$\text{Pass}, \quad \text{Not Pass}.$$

Let

$$p = \text{probability of passing}.$$

Then the probability of not passing must be

$$1 - p.$$

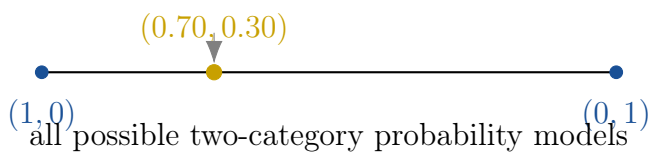
So the probability model is

$$(p, 1 - p).$$

There are two category probabilities, but there is only one free parameter. Once we choose p , the other probability is forced. For example,

$$(0.70, 0.30)$$

means a 70% chance of passing and a 30% chance of not passing. Geometrically, all possible two-category models lie on a line segment.



The left endpoint means all probability is in the first category. The right endpoint means all probability is in the second category. Points between the endpoints represent mixtures.

8.1.4 Three categories: models in a triangle

Now suppose there are three categories:

Pizza, Tacos, Salad.

A probability model might be

$$(0.50, 0.30, 0.20).$$

This means 50% Pizza, 30% Tacos, and 20% Salad. The probabilities must satisfy

$$p_1 + p_2 + p_3 = 1.$$

Because the total probability is fixed at 100%, choosing more of one category leaves less room for the others. For three categories, the possible probability models form a triangle.

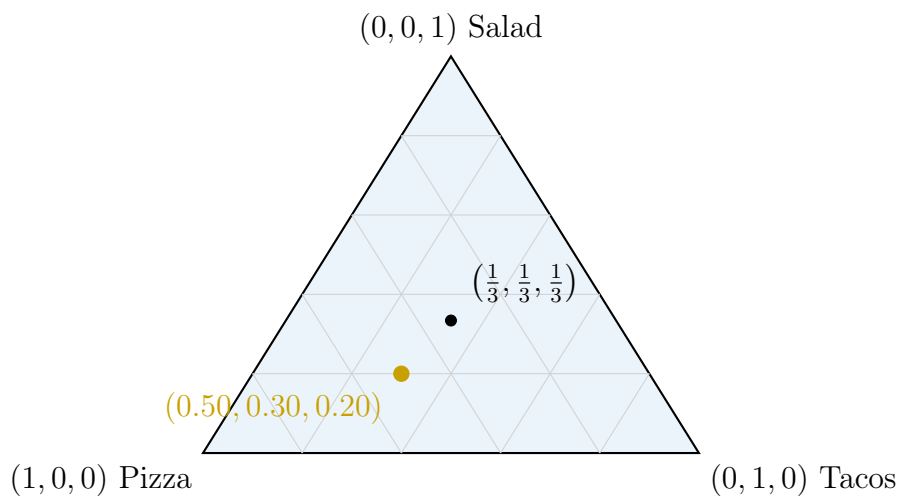


Figure 8.1: For three categories, the probability models form a triangle. Each point in the triangle is a possible mixture of the three categories.

Each corner represents putting all probability into one category. The center represents the equally likely model

$$\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Mathematicians call this triangle a **simplex**. In this class, the most important idea is that a probability model for three categories can be drawn as a point inside a triangle.

8.1.5 Four categories: models in a tetrahedron

With four categories, the same idea continues. Suppose the categories are

Pizza, Tacos, Salad, Sandwiches.

A probability model might be

$$(0.40, 0.25, 0.20, 0.15).$$

The probabilities must add to 1:

$$p_1 + p_2 + p_3 + p_4 = 1.$$

The shape is now a solid triangular shape called a **tetrahedron**. A tetrahedron has four corners. Each corner represents putting all probability into one category:

$$(1, 0, 0, 0), \quad (0, 1, 0, 0), \quad (0, 0, 1, 0), \quad (0, 0, 0, 1).$$

Points inside the tetrahedron represent mixtures of the four categories.

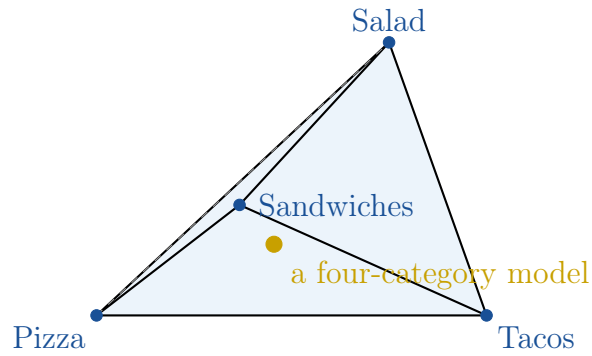


Figure 8.2: For four categories, the probability models form a tetrahedron.

The pattern is:

Number of categories	Probability constraint	Shape
2	$p_1 + p_2 = 1$	line segment
3	$p_1 + p_2 + p_3 = 1$	triangle
4	$p_1 + p_2 + p_3 + p_4 = 1$	tetrahedron

8.1.6 Free parameters and degrees of freedom

The geometry helps explain degrees of freedom. For two categories,

$$p_1 + p_2 = 1.$$

There are two probabilities, but only one free parameter. Once we choose p_1 , the other parameter is bound:

$$p_2 = 1 - p_1.$$

So the model has one free direction.

For three categories,

$$p_1 + p_2 + p_3 = 1.$$

There are three probabilities, but only two free parameters. Once we choose p_1 and p_2 , the third parameter is bound:

$$p_3 = 1 - p_1 - p_2.$$

So the model has two free directions.

For four categories,

$$p_1 + p_2 + p_3 + p_4 = 1.$$

There are four probabilities, but only three free parameters. Once we choose p_1, p_2, p_3 , the fourth parameter is bound:

$$p_4 = 1 - p_1 - p_2 - p_3.$$

So the model has three free directions.

This gives the goodness-of-fit degrees of freedom formula:

$$df = k - 1.$$

The “minus one” comes from the fact that the probabilities must add to 1.

Visual Insight. Degrees of freedom count how many independent directions the data can move after the constraints are accounted for. With k categories and a fixed total, one parameter is bound, so there are $k - 1$ free parameters.

8.1.7 Observed points, expected points, and chi-squared distance

Now return to data. Suppose 60 students choose among three lunch options and we observe

$$O = (30, 18, 12).$$

As proportions, this is

$$\left(\frac{30}{60}, \frac{18}{60}, \frac{12}{60}\right) = (0.50, 0.30, 0.20).$$

So the observed data gives a point inside the triangle.

In a chi-squared test, the null hypothesis gives us **expected counts**. For example, suppose the null hypothesis says students choose the three lunch options equally often. Then each option should get one-third of the total:

$$\frac{60}{3} = 20.$$

So the expected counts are

$$E = (20, 20, 20).$$

As proportions, this is

$$\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right),$$

which is the center of the triangle. Now we have two points: the observed point $O = (30, 18, 12)$ and the expected point $E = (20, 20, 20)$.

The chi-squared test asks whether the observed point is close to the expected point or far away.

The chi-squared statistic is

$$\chi^2 = \sum \frac{(O - E)^2}{E}.$$

For three categories, this means

$$\chi^2 = \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \frac{(O_3 - E_3)^2}{E_3}.$$

This is a distance formula, but it is not ordinary distance. It is a **weighted distance**. The expected counts determine the scale in each category.

A difference of 5 is a bigger deal when the expected count is 10 than when the expected count is 100:

$$\frac{(15 - 10)^2}{10} = 2.5,$$

while

$$\frac{(105 - 100)^2}{100} = 0.25.$$

Both have the same raw difference, but the first one contributes more to χ^2 because the expected count is smaller.

A useful way to see the geometry is to standardize each coordinate:

$$z_i = \frac{O_i - E_i}{\sqrt{E_i}}.$$

Then

$$\chi^2 = z_1^2 + z_2^2 + \cdots + z_k^2.$$

So χ^2 is the squared length of the standardized difference vector.

Visual Insight. The expected counts choose a target point in the simplex. They also determine how distance is measured. The chi-squared statistic is the squared weighted distance from the observed point to the expected point.

Example 8.1.1. (*Three lunch categories*) Suppose the null hypothesis says lunch choices are equally common among Pizza, Tacos, and Salad. A sample of 60 students gives

$$O = (30, 18, 12).$$

The expected counts are

$$E = (20, 20, 20).$$

Compute:

$$\chi^2 = \frac{(30 - 20)^2}{20} + \frac{(18 - 20)^2}{20} + \frac{(12 - 20)^2}{20}.$$

So

$$\chi^2 = \frac{100}{20} + \frac{4}{20} + \frac{64}{20} = 5 + 0.2 + 3.2 = 8.4.$$

The largest contributions came from Pizza and Salad. Geometrically, those categories are pulling the observed point away from the expected point.

Because there are three categories, the degrees of freedom are

$$df = 3 - 1 = 2.$$

The p-value tells us how unusual a chi-squared distance of 8.4 would be if the equal-choice model were true.

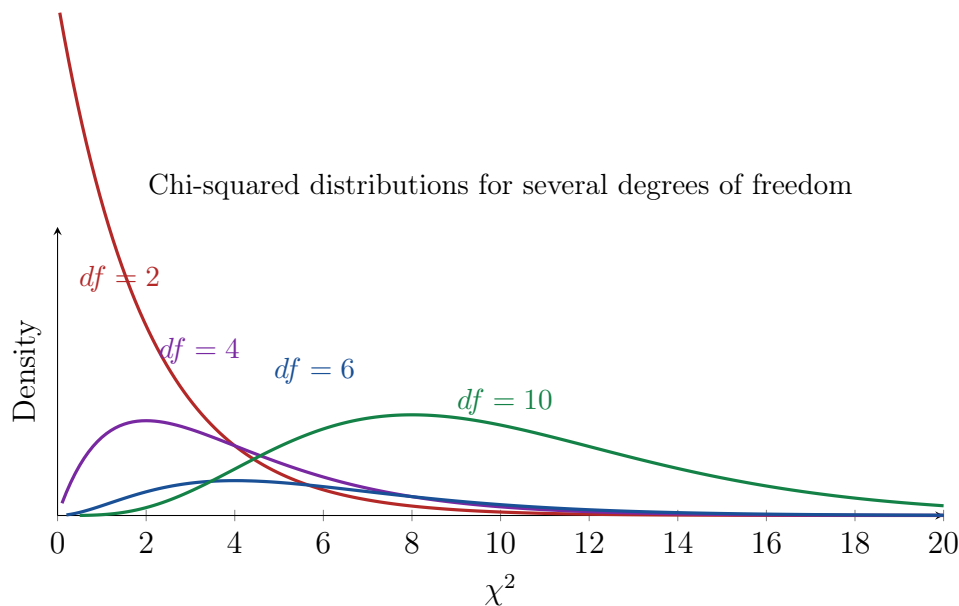
Visual Insight. Categorical models have geometry. Observed data gives a point in the shape. The null hypothesis gives an expected point in the same shape. A chi-squared goodness-of-fit test measures distance from that categorical model.

8.2 The Chi-Squared Distribution

Definition: Chi-Squared Distribution. The **chi-squared distribution** with k **degrees of freedom** (written χ_k^2) is a right-skewed continuous distribution defined on $[0, \infty)$. As k increases, the distribution becomes less skewed and more symmetric.

The number of **degrees of freedom** tells us which chi-squared curve to use. A chi-squared test with one degree of freedom uses a different curve from a test with five or ten degrees of freedom. In practice, the degrees of freedom come from the number of categories or the size of the two-way table.

Visual Insight. Unlike the normal distribution, the χ^2 distribution changes shape substantially with degrees of freedom. Small df gives a strongly right-skewed curve. Larger df gives a curve that is still non-negative, but less sharply skewed.



The p -value for a χ^2 test is always the area to the **right** of the test statistic. We do not split the area into two tails. A larger χ^2 value always means the observed counts are farther from the expected counts, so larger χ^2 values give stronger evidence against H_0 .

This right-tail behavior is one of the most important differences between chi-squared tests and the earlier z -tests. In a two-sided z -test, both very large positive and very large negative values count as evidence against H_0 . For chi-squared tests, there is no negative side. The statistic begins at zero and moves right as the mismatch grows.

The p -value is usually found using a chi-squared table. We find the row matching our degrees of freedom, then locate where our test statistic falls among the critical values. This lets us report conclusions such as $p < 0.05$, $p < 0.01$, or $0.01 < p < 0.05$.

8.3 The Chi-Squared Statistic: Observed vs. Expected

Before we introduce the formula, it helps to look at the kind of data these tests are designed for. Chi-squared tests are used when the data are **counts in categories**. We are not measuring a numerical variable like height, income, or test score. Instead, we are counting how many observations fall into each group.

For example, suppose a class surveys 100 students about their usual way of getting to campus. The raw data might look like a list of student responses:

Car Bus Car Walk Bike Car Bus Car Walk Car
 Bus Car Car Bike Walk Car Bus Car Car Walk
and so on, until all 100 responses are recorded

For a chi-squared test, we usually organize those raw responses into a frequency table. Each category gets one count.

Transportation type	Car	Bus	Walk	Bike	Other
Observed count	48	22	15	10	5

These numbers are the **observed counts**. They are what we actually saw in the sample. A null hypothesis gives us a competing picture of what we would expect to see. For instance, if someone claimed that the five categories should occur in the proportions 50%, 20%, 15%, 10%, and 5%, then in a sample of 100 students the expected counts would be 50, 20, 15, 10, and 5.

So the basic structure is:

Category	Car	Bus	Walk	Bike	Other
Observed count O	48	22	15	10	5
Expected count E	50	20	15	10	5

Each category is sometimes called a **cell** of the table. In this example, the Bus cell has observed count $O = 22$ and expected count $E = 20$. The chi-squared statistic measures the total disagreement between the observed row and the expected row, combining the discrepancies from all cells into one number.

Both chi-squared tests in this chapter use the same statistic. The idea is simple: compute what counts we would *expect* if H_0 were true, then measure how far the *observed* counts deviate from those expected counts.

Key Formula. The **chi-squared test statistic** is:

$$\chi^2 = \sum_{\text{all cells}} \frac{(O - E)^2}{E}$$

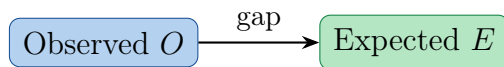
where O is the **observed count** in a cell and E is the **expected count** under H_0 .

Each term $\frac{(O-E)^2}{E}$ measures the squared relative discrepancy in one cell. We sum across all cells.

The formula has three pieces. First, $O - E$ measures the raw gap between what happened and what the null hypothesis predicted. Second, the square $(O - E)^2$ makes all gaps positive

and gives larger gaps more weight. Third, dividing by E adjusts for the size of the expected count. A gap of 5 is more surprising when the expected count is 10 than when the expected count is 500.

Visual Insight. Each term in the sum is a standardized squared gap:



$\frac{(O - E)^2}{E}$ is big when the gap is large relative to the expected count

If every observed count equals its expected count, then every term is zero and $\chi^2 = 0$. The data fit H_0 perfectly. As the data deviate from H_0 , χ^2 grows. Large χ^2 means **small** p -value and **more evidence against** H_0 .

Common Pitfall. The chi-squared test requires **counts**, not proportions or percentages. If you are given percentages, convert them to counts using the sample size before computing χ^2 .

Also: each expected count should satisfy $E \geq 5$. If any cell has $E < 5$, the chi-squared approximation may be unreliable, and the test should not be used as described here.

The condition about expected counts is easy to overlook because the observed counts are what we see in the data. But the approximation depends on the *expected* counts. A cell with a small expected count can make the chi-squared curve a poor approximation to the true sampling behavior of the test statistic.

8.4 Goodness-of-Fit Test

The **goodness-of-fit test** asks whether the observed distribution of one categorical variable matches a claimed distribution.

Examples:

- Does the distribution of M&M colors match the manufacturer's stated percentages?
- Are car accidents equally likely on all days of the week?
- Does the distribution of grades in a class match the expected distribution?

In a goodness-of-fit test, there is only one categorical variable. That variable has several possible categories. The null hypothesis gives the claimed proportion for each category, and we compare the observed category counts to the expected counts from that claim.

Key Formula. Chi-squared goodness-of-fit test:

H_0 : The population follows the claimed distribution.

H_a : The population does not follow the claimed distribution.

Expected counts: $E_i = n \cdot p_i$, where p_i is the claimed proportion for category i and n is the total sample size.

Test statistic: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

Degrees of freedom: $df = k - 1$, where k is the number of categories.

Condition: all expected counts $E_i \geq 5$.

The degrees of freedom are $k - 1$ because once the total sample size is fixed, not all category counts can vary freely. If you know the first $k - 1$ counts and the total count, the last count is forced.

Procedure. To run a goodness-of-fit test:

1. Write the claimed distribution as category probabilities.
2. Multiply each claimed probability by n to get expected counts.
3. Check that all expected counts are at least 5.
4. Compute $\sum (O - E)^2 / E$.
5. Use $df = k - 1$ to compare the statistic to a chi-squared critical value.

Example 8.4.1. (*Goodness-of-Fit: M&M Colors*) A candy company claims its bags contain 30% brown, 20% yellow, 20% red, 10% orange, 10% green, and 10% blue M&Ms. A student counts a bag of 200 M&Ms and gets:

Color	Brown	Yellow	Red	Orange	Green	Blue
Observed	65	35	42	18	24	16

Test whether the data are consistent with the company's claim at $\alpha = 0.05$.

Step 1: State the hypotheses and degrees of freedom.

H_0 : Colors follow the claimed distribution.

H_a : Colors do not follow the claimed distribution.

There are $k = 6$ color categories, so $df = k - 1 = 5$.

Step 2: Compute expected counts.

The expected counts come from $E_i = 200 \cdot p_i$. For example, the expected number of brown M&Ms is $200(0.30) = 60$, while the expected number of blue M&Ms is $200(0.10) = 20$.

Color	Brown	Yellow	Red	Orange	Green	Blue
p_i	0.30	0.20	0.20	0.10	0.10	0.10
E_i	60	40	40	20	20	20
O_i	65	35	42	18	24	16
$O - E$	+5	-5	+2	-2	+4	-4
$(O - E)^2/E$	0.417	0.625	0.100	0.200	0.800	0.800

All expected counts are at least 5, so the chi-squared condition is satisfied.

Step 3: Compute the test statistic.

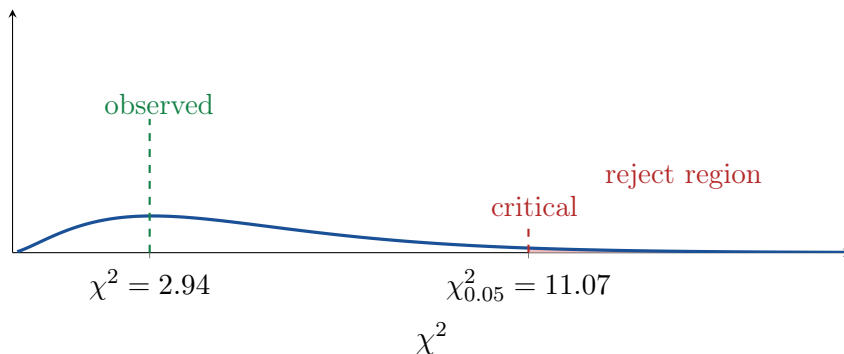
$$\chi^2 = 0.417 + 0.625 + 0.100 + 0.200 + 0.800 + 0.800 = 2.942.$$

This statistic is not measuring whether one color was high or low. It is measuring the total mismatch across all six colors.

Step 4: Use the chi-squared table and make a decision.

From the χ^2 table with $df = 5$, the critical value at $\alpha = 0.05$ is $\chi^2_{0.05,5} = 11.07$. Since $2.942 < 11.07$, the observed statistic is not in the rejection region. Thus $p > 0.05$, and we **fail to reject** H_0 .

Goodness-of-fit: $df = 5$, $p > 0.05$, fail to reject



Conclusion. There is insufficient evidence to conclude that the M&M color distribution differs from the company's claim. The bag's counts are consistent with the stated proportions. This does not prove the company's claim is exactly true; it only says this sample is not unusual enough to reject the claim at the 5% level.

Example 8.4.2. (*Goodness-of-Fit: Are Accidents Equally Likely by Day?*) Traffic records show 350 accidents at a particular intersection over one year:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
O_i	52	43	48	51	68	62	26

Test whether accidents are equally likely on all days (H_0 : each day has probability $1/7$) at $\alpha = 0.01$.

Step 1: State the hypotheses.

H_0 : Accidents are equally likely on all seven days, so $p_i = 1/7$ for each day.

H_a : Accidents are not equally likely on all seven days.

There are seven categories, so $df = 7 - 1 = 6$.

Step 2: Compute expected counts.

If accidents are equally likely on all seven days, then each day should have

$$E_i = \frac{350}{7} = 50$$

accidents.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
O_i	52	43	48	51	68	62	26
E_i	50	50	50	50	50	50	50
$(O - E)^2/E$	0.08	0.98	0.08	0.02	6.48	2.88	11.52

All expected counts are 50, so the expected-count condition is satisfied.

Step 3: Compute the test statistic.

$$\chi^2 = 0.08 + 0.98 + 0.08 + 0.02 + 6.48 + 2.88 + 11.52 = 22.04.$$

Step 4: Use the chi-squared table and make a decision.

The critical value at $\alpha = 0.01$ with $df = 6$ is $\chi_{0.01,6}^2 = 16.81$. Since $22.04 > 16.81$, we have $p < 0.01$ and **reject** H_0 .

Conclusion. Accidents are not equally distributed across days. Sunday's count (26) is much lower than expected, and Friday's count (68) is much higher than expected. Those two cells contribute a large part of the total chi-squared statistic, so they are important parts of the story.

Exercises 8.4

- 8.4.1. A die is rolled 120 times with results: 1→18, 2→22, 3→17, 4→25, 5→19, 6→19. Test whether the die is fair at $\alpha = 0.05$. ($df = 5$, critical value = 11.07.)
- 8.4.2. Compute the expected count for each cell in Exercise 1. What would it mean if every observed count exactly equaled its expected count?
- 8.4.3. The χ^2 statistic is always non-negative. Why? What is the minimum possible value and when does it occur?

8.5 Chi-Squared Test of Independence

The **test of independence** asks whether two categorical variables are related, or whether they behave as if they are independent of each other.

Examples:

- Is college major associated with political party preference?
- Is smoking status related to exercise frequency?
- Is a student's grade related to whether they attended tutoring?

Data for this test are arranged in a **two-way contingency table**. The rows represent one categorical variable, and the columns represent another categorical variable. Each cell contains the number of individuals who fall into that combination of row and column categories.

Definition: Independence. Two categorical variables are **independent** if knowing the value of one gives no information about the likely value of the other. If they are not independent, we say they are **associated**.

For example, if exercise frequency and sleep quality were independent, then the proportion of people reporting good sleep should be about the same among people who rarely exercise, sometimes exercise, and often exercise. If those proportions are very different, that suggests association.

Key Formula. Chi-squared test of independence:

H_0 : The two variables are independent (not associated).

H_a : The two variables are associated.

Expected count for the cell in row r , column c :

$$E_{rc} = \frac{(\text{row } r \text{ total}) \times (\text{column } c \text{ total})}{n}$$

Test statistic: $\chi^2 = \sum_{\text{all cells}} \frac{(O - E)^2}{E}$

Degrees of freedom: $df = (r - 1)(c - 1)$, where r = number of rows, c = number of columns.

Condition: all expected counts $E_{rc} \geq 5$.

8.5.1 Why That Expected Count Formula?

If two variables are truly independent, then the probability of being in row r *and* column c should equal the probability of row r times the probability of column c . The expected count is just n times that joint probability:

$$E_{rc} = n \cdot \frac{\text{row total}}{n} \cdot \frac{\text{col total}}{n} = \frac{\text{row total} \times \text{col total}}{n}.$$

If the observed counts match these expected counts, the variables behave as if independent. If they differ substantially, the variables are associated.

A useful way to remember this is: the row total tells us how many people are in a row category, and the column proportion tells us how those people would be split among the columns if there were no association. Multiplying row total by column proportion gives the expected count.

Procedure. To run a test of independence:

1. Make a two-way table with row totals, column totals, and grand total n .
2. Compute each expected count using $E = (\text{row total})(\text{column total})/n$.
3. Check that all expected counts are at least 5.
4. Compute $(O - E)^2/E$ for every cell and add the results.
5. Use $df = (r - 1)(c - 1)$ to compare the statistic to a chi-squared critical value.

Example 8.5.1. (*Test of Independence: Exercise and Sleep*) A health researcher surveys 300 adults about their exercise frequency and sleep quality. Results:

Exercise	Sleep Quality			Row total
	Poor	Fair	Good	
Rarely	42	38	20	100
Sometimes	28	40	32	100
Often	10	32	58	100
Col total	80	110	110	300

Test whether exercise frequency and sleep quality are independent at $\alpha = 0.05$.

Step 1: State the hypotheses and degrees of freedom.

H_0 : Exercise frequency and sleep quality are independent.

H_a : Exercise frequency and sleep quality are associated.

The table has 3 rows and 3 columns, so

$$df = (3 - 1)(3 - 1) = 4.$$

Step 2: Compute expected counts.

Use $E = (\text{row total})(\text{column total})/n$. For the Rarely/Poor cell,

$$E = \frac{100 \cdot 80}{300} = 26.67.$$

For the Rarely/Good cell,

$$E = \frac{100 \cdot 110}{300} = 36.67.$$

Because all three row totals are 100 in this example, each row has the same set of expected counts.

	Poor	Fair	Good
Rarely	26.67	36.67	36.67
Sometimes	26.67	36.67	36.67
Often	26.67	36.67	36.67

All $E \geq 5$. ✓

Step 3: Compute $(O - E)^2/E$ for each cell and sum.

	Poor	Fair	Good
Rarely	$(42 - 26.67)^2/26.67 = 8.83$	$(38 - 36.67)^2/36.67 = 0.05$	$(20 - 36.67)^2/36.67 = 7.59$
Sometimes	$(28 - 26.67)^2/26.67 = 0.07$	$(40 - 36.67)^2/36.67 = 0.30$	$(32 - 36.67)^2/36.67 = 0.59$
Often	$(10 - 26.67)^2/26.67 = 10.42$	$(32 - 36.67)^2/36.67 = 0.59$	$(58 - 36.67)^2/36.67 = 12.42$

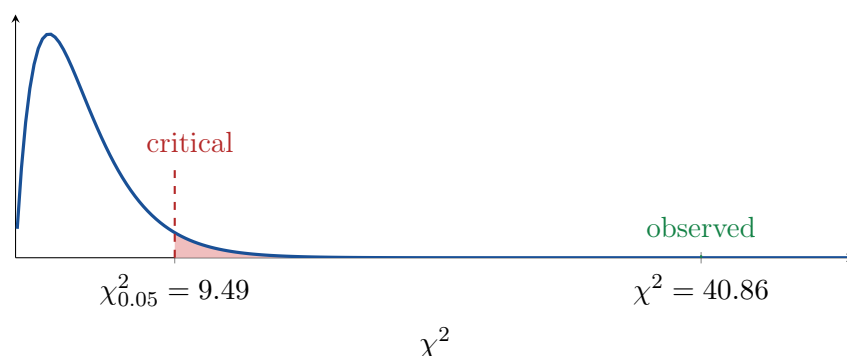
$$\chi^2 = 8.83 + 0.05 + 7.59 + 0.07 + 0.30 + 0.59 + 10.42 + 0.59 + 12.42 = 40.86.$$

The largest contributions come from cells where the observed count is far from what independence would predict. Those cells help us interpret the association, not just decide whether it exists.

Step 4: Use the chi-squared table and make a decision.

The critical value at $\alpha = 0.05$ with $df = 4$ is $\chi^2_{0.05,4} = 9.488$. Since $40.86 \gg 9.488$, the statistic is far into the right tail, so $p \ll 0.001$. We **reject** H_0 .

Test of independence: $df = 4$, $p \ll 0.001$, reject H_0



Conclusion. There is very strong evidence ($p \ll 0.001$) that exercise frequency and sleep quality are associated. People who exercise often are much more likely to report good sleep, while those who rarely exercise are much more likely to report poor sleep. The cells with the largest contributions to χ^2 are “Rarely/Poor” (8.83), “Rarely/Good” (7.59), “Often/Poor” (10.42), and “Often/Good” (12.42) - the corners of the table. These cells show that the association runs in the expected direction.

Common Pitfall. Association is not causation. Rejecting H_0 in a chi-squared test of independence tells us the two variables are statistically related. It does *not* tell us that one causes the other. People who exercise more might also have less stress, better diet, or different genetics - any of which could explain better sleep. Chi-squared measures association; causal claims require experimental design.

Exercises 8.5

8.5.1. A survey asks 200 college students about their class year (Freshman, Sophomore, Junior, Senior) and preferred study location (Library, Dorm, Cafe). Set up the hypotheses and state the degrees of freedom.

8.5.2. For a 4×3 contingency table with grand total $n = 500$, a cell has row total 120 and column total 75. Compute its expected count.

8.5.3. In the sleep/exercise example, identify the two cells that contribute the most to χ^2 . What does each tell you about the association?

8.6 Reading a Chi-Squared Table

Since we present χ^2 results using a critical value table rather than computing exact p -values, we need to be comfortable reading the table. The table gives right-tail critical values. A critical value is a cutoff: statistics to the right of the cutoff are unusual enough to reject the null hypothesis at that significance level.

A typical chi-squared table lists critical values by df (rows) and significance level α (columns).

df	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.001$
1	2.706	3.841	6.635	10.828
2	4.605	5.991	9.210	13.816
3	6.251	7.815	11.345	16.266
4	7.779	9.488	13.277	18.467
5	9.236	11.070	15.086	20.515
6	10.645	12.592	16.812	22.458
7	12.017	14.067	18.475	24.322
8	13.362	15.507	20.090	26.124
9	14.684	16.919	21.666	27.877
10	15.987	18.307	23.209	29.588

How to use the table:

1. Identify df and locate the correct row.
2. Compare your χ^2 statistic to the critical values in that row.
3. If $\chi^2 > \chi^2_{\alpha, df}$, then $p < \alpha$ and we reject H_0 at level α .
4. If $\chi^2 < \chi^2_{\alpha, df}$, then $p > \alpha$ and we fail to reject at level α .

You can also bracket the exact p -value. For example, if $df = 4$ and $\chi^2 = 11.2$, then the statistic exceeds 9.488 ($\alpha = 0.05$) but does not exceed 13.277 ($\alpha = 0.01$). Therefore,

$$0.01 < p < 0.05.$$

This is often enough information to make a conclusion.

Exercises 8.6

8.6.1. For each χ^2 statistic and df , report the conclusion at $\alpha = 0.05$ and bracket the p -value as best you can using the table. (a) $\chi^2 = 7.1$, $df = 3$. (b) $\chi^2 = 18.5$, $df = 6$. (c) $\chi^2 = 4.2$, $df = 2$.

8.6.2. What happens to critical values as df increases (for fixed α)? Why does this make intuitive sense?

8.7 Comparing the Tests: A Growing Toolkit

We now have six hypothesis tests. The chi-squared tests occupy a distinct corner: they apply to *categorical* data where the question involves *counts* across categories, not means or proportions of a single outcome.

Test	Data type	Question
One-sample z for μ	Quantitative	Is μ equal to a claim?
One-sample z for p	Categorical (binary)	Is p equal to a claim?
Two-sample z for $\mu_1 - \mu_2$	Quantitative	Do two groups have the same μ ?
Two-sample z for $p_1 - p_2$	Categorical (binary)	Do two groups have the same p ?
χ^2 goodness-of-fit	Categorical (multiple)	Does this distribution fit a claim?
χ^2 test of independence	Two categorical vars.	Are these variables associated?

The most common mistake is to choose a test based on the words in the problem instead of the structure of the data. Before choosing a test, ask two questions:

1. Is the response quantitative or categorical?
2. Are we studying one variable, two groups, or two categorical variables in a table?

If the data are categorical counts spread across several categories, a chi-squared test is often the right tool. If the data are numerical measurements, we usually need a test about means instead.

Chapter 8 Review

1. A botanist claims seeds from four plant varieties germinate in the ratio 4:3:2:1. She plants 200 seeds and observes: Variety A (88), Variety B (60), Variety C (35), Variety D (17). (a) State H_0 and H_a . (b) Compute expected counts. (c) Compute χ^2 . (d) State the conclusion at $\alpha = 0.05$ ($df = 3$, critical value 7.815).

2. A survey of 400 employees at a company records gender (Male/Female) and preferred work arrangement (Remote/Hybrid/Office):

	Remote	Hybrid	Office	Total
Male	80	60	40	180
Female	90	70	60	220
Total	170	130	100	400

- (a) State H_0 and H_a . Identify df . (b) Compute all six expected counts. (c) Compute χ^2 . (d) State the conclusion at $\alpha = 0.05$ (critical value 5.991).
3. A student computes $\chi^2 = 3.2$ with $df = 1$. (a) Is this significant at $\alpha = 0.05$? At $\alpha = 0.10$? (b) What is the minimum χ^2 value needed to reject H_0 at $\alpha = 0.01$ with $df = 1$?
4. A study finds a significant association between ice cream sales and drowning deaths ($\chi^2 = 18.4$, $df = 1$, $p < 0.001$). A journalist writes “Ice cream causes drowning.” What is wrong with this conclusion? What confounding variable might explain the association?
5. *Big picture*: Compare the chi-squared goodness-of-fit test to the one-sample z -test for a proportion. Both test whether data match a claimed distribution. When would you use each? What does the goodness-of-fit test offer that the one-sample proportion test cannot?